Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Based on what you have learned this week, what is the best way to describe a Kalman filter? — 1 / 1 point

- ( ) As an analog circuit using operational amplifiers and passive components to filter noise out of a signal.
- ( ) As an electronic device incorporating microcontrollers and transistors.
- ( ) As a PC-based program incorporating a graphical user interface and internet communications.
- (•) As an algorithm usually implemented as a computer program, designed to estimate the state of a dynamic system.

> **Correct**
> Yes, this is the purpose and typical way to implement a Kalman filter.

2. Based on what you have learned this week, which of the following applications might be based on a Kalman filter? (Select all that apply.) — 1 / 1 point

- [x] A parameter estimator

> **Correct**
> Yes, parameter-estimation algorithms often use Kalman filters. You will learn more about this in the third course in the specialization.

- [x] A target-tracking system

> **Correct**
> Yes, target-tracking systems often use Kalman filters. You will learn more about this in the second course in the specialization.

- [x] A state estimator

> **Correct**
> Yes, state-estimation algorithms often use Kalman filters. You will learn more about this in the fourth week of this course.

- [x] A navigation system

> **Correct**
> Yes, navigation systems often use Kalman filters. You will learn more about this in the fourth course in the specialization.

3. In the discrete-time state-space model: — 1 / 1 point

$$x_{k+1} = Ax_k + Bu_k + w_k$$

$$z_k = Cx_k + Du_k + v_k,$$

which of the following is true?

- ( ) x_k is the system's state vector, u_k is the deterministic input, w_k is sensor noise, z_k is a sensor measurement, and v_k is the disturbance or process noise input.
- ( ) x_k is the system's state vector, u_k is a sensor measurement, w_k is the disturbance or process noise input, z_k is the deterministic input, and v_k is sensor noise.
- (•) x_k is the system's state vector, u_k is the deterministic input, w_k is the disturbance or process noise input, z_k is a sensor measurement, and v_k is sensor noise.
- ( ) x_k is the deterministic input, u_k is the system's state vector, w_k is the disturbance or process noise input, z_k is a sensor measurement, and v_k is sensor noise.

> **Correct**
> Yes, you are correct.

4. If our goal is to make an optimal estimate of x_k in the discrete-time state-space model: — 1 / 1 point

$$x_{k+1} = Ax_k + Bu_k + w_k$$

$$z_k = Cx_k + Du_k + v_k,$$

why do we need feedback? (Select all that apply.)

- [x] Because the values of the disturbance signal w_k are not known, and feedback helps to correct for the uncertainty that this contributes to the x_k signal.

> **Correct**
> Yes. This is one reason that we need feedback.

- [x] Because we never know the values for the model's matrices A, B, C and/or D perfectly, and feedback helps to correct for errors in our model.

> **Correct**
> Yes. This is one reason that we need feedback.

- [x] Because we rarely (if ever) know the initial state x_0 exactly and feedback helps to correct for an uncertain guess of x_0 over time.

> **Correct**
> Yes. This is one reason that we need feedback.

- [ ] Because we can never measure the u_k signal and must use feedback to help correct for the uncertainty that is caused by this inability.

5. What is the operating principle of a discrete-time Kalman filter? — 1 / 1 point

- ( ) Based on an initial state estimate $\hat{x}_0$, the algorithm repeatedly predicts process noise w_{k-1} and creates a state estimate by computing $\hat{x}_k = A\hat{x}_{k-1} + Bu_{k-1} + w_{k-1}$.
- ( ) Based on an initial state estimate $\hat{x}_0$, the algorithm then predicts the value of $\hat{x}_1^-$ and corrects that value to $\hat{x}_1^+$ after measuring z_1. Then, the algorithm runs open-loop to make all other estimates $\hat{x}_k^+$.
- ( ) Based on an initial state estimate $\hat{x}_0$, the algorithm repeatedly creates a state estimate by computing $\hat{x}_k = A\hat{x}_{k-1} + Bu_{k-1}$.
- (•) Based on an initial state estimate $\hat{x}_0$, the algorithm then predicts the value of $\hat{x}_1^-$ and corrects that value to $\hat{x}_1^+$ after measuring z_1. Then, the algorithm repeatedly executes prediction and correction (estimation) steps for every measurement z_k.

> **Correct**
> Yes, this is correct.

6. What information do we need in order to design a Kalman filter? (Select all that apply.) — 1 / 1 point

- [x] We need a discrete-time state-space model of the dynamics of the system.

> **Correct**
> Yes, this is true.

- [x] We need to describe what we know about the uncertain process noise w_k and sensor noise v_k.

> **Correct**
> Yes, this is true.

- [x] We need to describe what we know about the uncertain initial state x_0 of the system.

> **Correct**
> Yes, this is true.

- [ ] We need to have manufacturer part numbers for the sensors that will be used with the Kalman filter.

7. What is the full output of a Kalman filter at any given timestep? — 1 / 1 point

- (•) The estimated value of the state vector at this timestep, plus confidence bounds on this estimate.
- ( ) The predicted value of the state vector at this timestep, plus confidence bounds on this prediction.
- ( ) The estimated value of the state vector at this timestep.
- ( ) The predicted value of the state vector at this timestep.

> **Correct**
> Yes. This is correct.

8. What background topics are essential to being able to derive and understand Kalman filters? (Select all that apply.) — 1 / 1 point

- [x] Probability theory and stochastic systems.

> **Correct**
> Yes, which is why we study them in week 3 of this course.

- [ ] Artificial intelligence.
- [x] Least-squares estimation theory.

> **Correct**
> Yes, which is why we study this topic in the second course.

- [x] State-space dynamic systems.

> **Correct**
> Yes, which is why we study them next week.

9. What is sequential probabilistic inference? — 1 / 1 point

- ( ) A special type of state estimator that applies only to linear systems. The present state estimate is computed afresh every timestep based on the entire history of input and output measurements.
- (•) A general type of state estimator, of which the Kalman filter is a special case. The present state estimate is computed recursively based on the prior estimate and new input and output measurements.
- ( ) A special type of state estimator that applies only to linear systems. The present state estimate is computed recursively based on the prior estimate and new input and output measurements.
- ( ) A general type of state estimator, of which the Kalman filter is a special case. The present state estimate is computed afresh every timestep based on the entire history of input and output measurements.

> **Correct**
> Yes, this is correct.

10. What is the principal reason why it is important to study how the Kalman filter is derived? — 1 / 1 point

- ( ) Because it is valuable to see how the Kalman filter solved the problem of optimal state estimation, for historic perspective and interest.
- (•) Because many applications violate the assumptions made in the derivation and require that we are able to make modifications to the derivation.
- ( ) Because it is a good opportunity to practice and refine math skills.
- ( ) Because Kalman filters are not actually used for state estimation and must be modified so that they can be used for other applications.

> **Correct**
> Yes. This is correct.